Discover Artificial Intelligence

Research

Automatic two-wheeler rider identification and triple-riding detection in surveillance systems using deep-learning models

Uttam U. Deshpande[1] · Supriya Shanbhag[1] · Rudragoud Patil[2] · Ramchandra Alias Ameet Chate[3] · Sufola Das Chagas Silva Araujo[4] · Kevin Pinto[1] · J. Arjun[1] · Mayankraj Patil[1] · Balagouda Nirwani[1]

Received: 1 January 2025 / Accepted: 10 April 2025
Published online: 15 June 2025

Abstract
Uncontrolled road traffic conditions are commonly seen in South Asian countries, which result in the majority of motorcycle accidents due to triple riding, and helmetless driving traffic violation incidents. Triple riding is a dangerous act that can result in serious legal consequences. Each rider should be aware of the stringent traffic safety regulations of helmet wear and triple-riding violations. Public safety can be improved by reducing the number of road accidents. To do this, these riders must be identified and prosecuted. With little assistance from humans, the automated traffic monitoring system can enforce rigorous adherence to traffic laws. The current methods are effective when applied to widely used datasets, like Kaggle and COCO, which offer a helpful research platform. However, it is difficult to obtain satisfactory detection accuracies because this dataset contains minimal triple-riding images and lacks the sensation of realistic traffic CCTV images obtained from specific heights and angles. We provide a real-time solution that employs surveillance cameras placed at various angles and heights to detect two-wheelers, identify the number of riders, and recognize the vehicle involved in this traffic violation. To address challenging environments like occlusions and precise vehicle detection from a long distance we use the ResNet18-based DetectNet_v2 model. To reliably predict triple riding from several riders sitting on a two-wheeler and extract license plate information, we employ a cutting-edge YOLOv8 object-detection algorithm that operates on the Darknet framework. After experiment analysis, we found that our proposed model demonstrated a promising triple-rider, two-wheeler, and numberplate detection accuracy of 91.42%, 98%, and 81% respectively under challenging situations.

Keywords Triple riding · Kaggle · COCO · ResNet18 · DetectNet_v2 · YOLOv8 · Darknet

1 Introduction

Irrespective of demographic or geographic boundaries, road safety is a major concern across the world. Road transportation, especially on two-wheelers, is the most economical of transportation in densely populated nations like India. Rapid urbanization and an unexpected motorization rate are leading to poor traffic management, which in turn

✉ Uttam U. Deshpande, uttamudeshpande@gmail.com; Supriya Shanbhag, ssshnbhag@git.edu; Rudragoud Patil, rspatil@git.edu; Ramchandra Alias Ameet Chate, ameet.chate@gmail.com; Sufola Das Chagas Silva Araujo, sufola@pccegoa.edu.in; Kevin Pinto, kevinvmp17@gmail.com; J. Arjun, arjunjamkhandi2002@gmail.com; Mayankraj Patil, mayankrajpatil@gmail.com; Balagouda Nirwani, balagoudanirwani22@gmail.com | [1]Department of Electronics and Communication Engineering, KLS Gogte Institute of Technology, Belgaum, Karnataka, India. [2]Department of Computer Science and Engineering, KLS Gogte Institute of Technology, Belgaum, Karnataka, India. [3]Department of MBA, KLS Gogte Institute of Technology, Belgaum, Karnataka, India. [4]Department of Computer Science and Engineering, Padre Conceição College of Engineering, Goa, India.

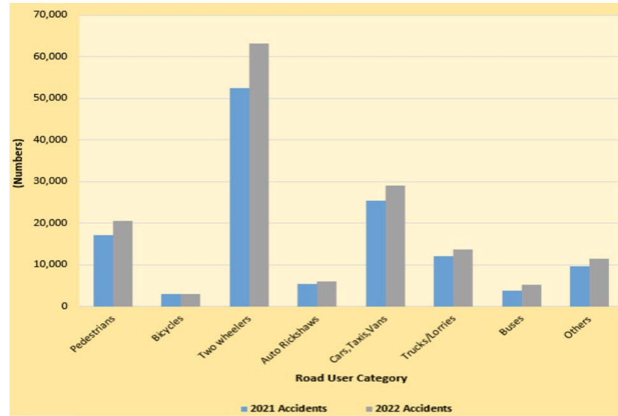


Fig. 1 Accidents were reported under different vehicle categories during 2021–22

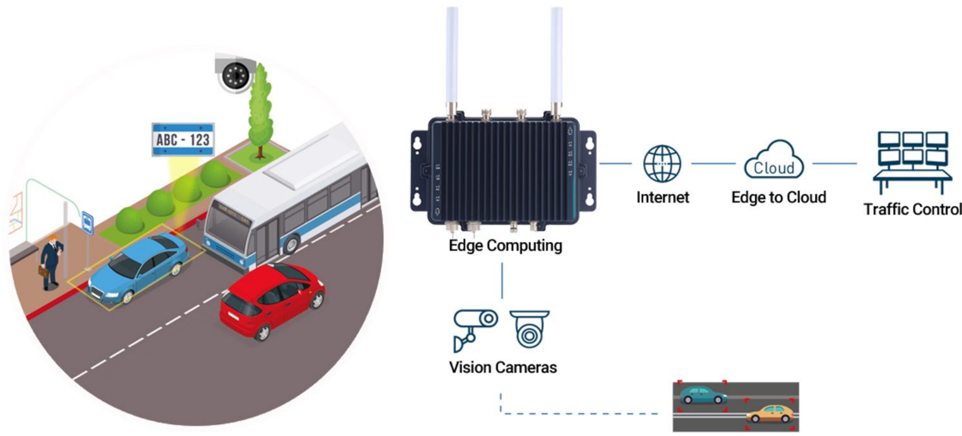


Fig. 2 Automatic traffic law enforcement in ITS

causes traffic accidents, injuries, and fatalities. From 2010 to 2020, the overall number of motor vehicle registrations increased by 9.8%, according to statistics from the Indian Road Transport Department. As of 2020, two-wheelers made up 74.7% of all vehicles, including cars, taxis, jeeps, buses, and freight vehicles. In 2022, two-wheelers were also responsible for the highest number of fatalities (25,228) and accidents (63,115) [1] as indicated in Fig. 1.

In 2022, about 50 thousand people were killed (71% riders and 29% pillion riders), and more than 1lak people got injured in road accidents for ignoring to wear helmets [1]. In the same year, Delhi accounted for more than 44% of pedestrian fatalities in traffic incidents involving reckless and triple-riding incidents. The traffic police prosecuted more than 19,000 two-wheeler triple-riding cases [2]. Given the daily rise in two-wheeler infractions, manual traffic monitoring presents several challenges for preserving public road safety and strict adherence to traffic laws. Triple-riding and disobeying traffic signals are very dangerous practices. Public safety is seriously endangered by these traffic infractions and careless driving. The effectiveness of manual enforcement techniques is limited due to limited human resources as well as errors that are introduced due to manual interventions. Therefore, an automated system that employs deep learning and computer vision techniques can provide an accurate and efficient means to identify and enforce stringent traffic infraction laws. The need for automated traffic monitoring and surveillance systems [3–5] is therefore steadily increasing. The urban growth has greatly benefited from Intelligent Transportation Systems (ITS). ITS employs a range of techniques, including edge-AI cameras, sensors, communication devices, etc., to provide a quick, intelligent, and reliable way to identify helmet, triple riding, and traffic signal infractions as illustrated in Fig. 2. To figure out traffic flow, vehicle speed, congestion, and traffic violations in real time, traffic managers analyze the data sent by traffic Internet of Thongs (IoT) sensors, cameras, and GPS devices. Advanced embedded edge Artificial Intelligence (AI) hardware, AI algorithms, smart cameras, and IoT cloud can be used to automate police enforcement and traffic control. It can record events in real time and use analysis to derive important insights. It offers a practical method to cut down on traffic infractions while utilizing fewer police resources.

To recognize basic shapes, determine shading differences, and locate contours and colors, traditional object detection techniques rely on manually annotated features and heuristic algorithms. When the dataset is limited and a small improvement in detection accuracy is anticipated, this approach performs well in triple-riding violation detections [6–8] as illustrated in Fig. 3.

Triple-riding incidents observed in Fig. 3a, b do not represent realistic traffic scene images captured from far distances and heights. Apart from this, many incidents involving pedestrians falsely detected as riders (Fig. 3a) are reported due to a lack of reliable riders on a two-wheeler information extraction. Furthermore, occlusions, cluttered backgrounds, and far-distance traffic images are not robustly handled by these approaches. To overcome these problems, we employ the bounding-box regression method known as "GridBox-object detection" from the DetectNet_v2 (ResNet18) architecture [9]. This makes it possible to reliably extract information related to riders seated on two-wheelers from long distances and reduce False Positives. Most existing studies on triple rider detection classify riders based on their predicted face information [10–12]. This leads to performance issues in crowded road scenarios due to missing Regions of Interest (ROI) information. To increase detection accuracy, we use the ROIs of two-wheelers to determine the number of riders and triple-riding violations. We achieve this through an automated deep learning-based YOLOv8 model. We evaluate the performance of the proposed model and classify two-wheelers with and without triple-riding classes. We created the data set from cameras installed at various locations in the college and on the main roads. Our proposed system detects triple-riders with an accuracy of 91.42%.

The following is a summary of our novel contributions:

1. A diverse dataset containing 6564 images was created by acquiring videos from wide dynamic range cameras (PTZ, bullet) mounted at various angles and heights (12–15 feet) of our college campus and main streets.
2. A cutting-edge transfer-learning-based dataset pre-processing pipeline containing a bounding box regressor ResNet18-based DetectNet_v2 was utilized to generate two-wheeler boundary representations.
3. A YOLOv8 model was deployed to identify the number of riders, detect triple-riding violations, and extract vehicle information from complex traffic environments. The findings show that the model effectively detects multiple objects from large field-of-view traffic scenes.

The paper is organized in the following manner: The related work is discussed in Sect. 2, a triple riding detection system is proposed in Sect. 3, performance is assessed in Sect. 4, and the results, the limitations of the suggested technique, and the future scope are discussed in Sect. 5.

2 Related works

Digital image processing and computer vision are utilized in many application fields, including autonomous driving, pose estimation, robotics, and surveillance systems [13–16]. Research focus has recently shifted to deep learning model development, which has produced positive outcomes across a range of applications [17–19]. The machine learning models receive hand-crafted features as input, and objects are classified using methods such as the supper vector machine

(a)

(b)

Fig. 3 Limitations of the existing object detection methods **a** Pedestrian wrongly detected as a rider by YOLOv8 model due to inaccurate feature extraction [6], **b** triple riding dataset not representing real-time traffic scenario [7]

(SVM). The feature extraction and mapping process determines the overall accuracy of the network; for deeper feature mappings, this may involve stacking many convolutional and pooling layers. VGGNet [20], ResNet [21], GoogleNet [22], and AlexNet [23] are a few popular feature extraction architectures. Since its initial release in 2012, AlexNet has primarily been utilized for image classification applications. It has 3 pooling, 3 fully connected, and 5 convolutional layers. VGGNet created other variations with extra layers, such as VGG-16/19, to improve performance and internally fortify the design. While ResNet established the notion of skip-connections to preserve and make information available across system layers, GoogleNet introduced the cascade idea known as "inception" modules.

An object-detector's job is to determine whether a relevant object is present in the image of a video frame. If that object is present, the detector provides the appropriate class, locality, and object's dimensions. Deep learning-based methods may identify and classify objects using either a single-stage or two-stage CNN [24]. The two primary responsibilities of object detectors are to identify objects with their bounding boxes) and then classify them after extracting features from the input image. The object region is first obtained in a two-stage architecture, and classification is then done using the features that were captured out of the suggested region. These systems can achieve extremely high accuracy rates, but they are not suited for real-time scenarios like autonomous vehicles due to their poor speed. The object detector based on their classification is illustrated in Fig. 4.

R-CNN [25], Fast R-CNN [26], Faster R-CNN [27], RF-CN [28], and Mask R-CNN [29] are examples of 2-stage detection techniques. These algorithms are sometimes referred to as "region-based" since they aim to ensure the object's position inside the image by creating a single-shot bounding box around its region of interest. Single-stage detection methods estimate the objects' existence and location by only feeding the input image into the network once. Because these methods process the incoming data in a single-pass, they are computationally efficient. The single-shot detection algorithm's drawback is its lower predictive accuracy, making it best suited for real-time surveillance in environments with constrained resources.

2.1 One-stage YOLO object detector

Single-stage detection techniques estimate the existence and location of the objects by just passing the input picture into the network once. These techniques are computationally efficient since they evaluate the given image in a single pass. The single-shot detection algorithm's drawback is its lower predictive accuracy, making it best suited for real-time surveillance in environments with constrained resources. Single-stage detection techniques are computationally efficient since they determine the identity and location of the objects by just feeding the input image into the network once. In the year 2016, Ali Farhadi et al. developed the first variant of the YOLO model [30]. As previously stated, YOLO is a CNN-based one-step detection architecture that uses the Single-Shot Multi-box Detector (SSD) single-pass technique for object identification and classification tasks [31]. A multi-box approach is used in boundary box regression. The detector network consists of two stages: feature-map extraction and object-detection using CNN filters. SSD characteristics are extracted using VGG 16. Numerous models have been released since YOLO was first introduced in 2015. The third variation of the YOLO model was published in 2018. A neural network post-processing approach and a full CNN comprise the

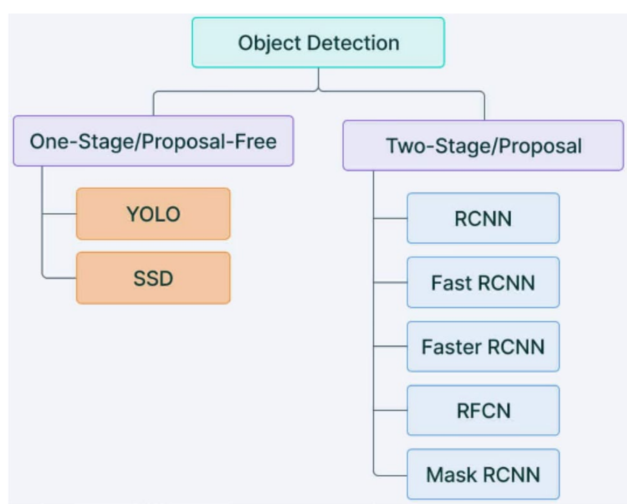


Fig. 4 Object detector classifier examples

YOLO model. Consequently, the method employed depends on the kind of application for which the object-detecting system is designed. Many optimization strategies based on machine learning have been published in the literature [32]. A specifically improved variant of ResNet called "Cross-Stage Partial Network (CSPNET)" was created specifically for object detection tasks. This new architecture with as low as 53 layers, served as the foundation for the YOLO v4 model [33].

YOLOv5 the next variant called "EfficientDet" uses a denser backbone network design and unlike previous versions, the model can be trained on multiclass objects. Models' improvement in recognizing objects' varying shapes was achieved by adding Spatial Pyramid pooling layers. EfcientNet-L2 served as the foundation for the new CNN-based architecture known as YOLO v6, which exceeded previous versions regarding accuracy and processing efficiency. The YOLO v7 utilizes seven anchor boxes [34] to identify objects with various aspect ratios and shapes. To effectively detect small objects, a revised model with a "focal loss" function was added. Compared to previous model versions, this model is the most efficient in terms of detection accuracy, and speed. The newly released YOLO v8 version performs better than its predecessors on the popular COCO Dataset, demonstrating improved precision and versatility in object detection tasks. On the popular COCO Dataset, the recently released YOLO v8 version outperforms its predecessors, exhibiting improved accuracy and versatility in object detection applications. A standard Python library with a command-line interface for training, object detection, and model prediction are some of the key features. The anchor-free feature makes it easier to predict an object's center rather than its offset concerning the anchor boxes. Thus, this model's detection is more effective. Many researchers have utilized these models in their rider helmets and triple riding detection which are discussed next.

2.2 Vehicle, rider helmet, and triple riding detection

In 2018, the researchers provided an in-depth review of vehicle and pedestrian detection [35]. The study examined the many techniques used in the literature to identify vehicles and humans. Based on the survey's findings they concluded that deep learning is the most accurate technique for recognizing and predicting objects in traffic. To detect helmet violations, the RetinaNet and RestNet-50 deep learning models were used [36]. They encountered setbacks in carrying out research due to varying camera angles, and atmospheric conditions affecting the image annotation job. The YOLO v3 model was utilized [37] to identify the motorcycle and license plates. The suggested method achieves 89% and 92% accuracy for motorcycle and number plate detection, respectively, on two deep-learning models for the tasks. On the KITTI dataset, a transfer learning-based (YOLOv3) vehicle identification system [38] outperformed the earlier methods in terms of detection accuracy. The models designed for helmet detection are trained on clear camera footage [39] and they cannot identify traffic violations in cameras placed at great heights or distances. The work carried out using foreground segmentation [40] followed by ROI-based rider helmet detection [10, 41–43] produced robust performance on clear images. One of the many drawbacks of CNN-based helmet/mask identification and vehicle categorization [44] is that it incorrectly identifies the helmet as a scarf. As a result, accuracy is low. A YOLOv5 model trained using ensemble-learning [45] for a real-time helmet violation detection system showed improvement in the motorcycle riders and helmet wear detections. During the ensemble-learning phase, five models with different hyperparameters were set. The model was able to attain a mAP score of 0.5267, paving the way for improvement in the precision and recall values. Nevertheless, these works do not address triple-riding detection problems.

Very few techniques have been devised to address triple-riding incidents. To identify, detect, and report helmet and triple rider offenses on unconstrained road scenarios, an amodal regressor and curriculum-based learning strategy [46] using boundary boxes was employed. Nearly 90% accuracy for helmet detection and 86% accuracy for motorbike and rider detection were reported by this method. The technique developed by Nikhil et al. [7] uses the information gathered from the lower part of the motorcycle image to identify violations. The model has a generalization issue and is prone to false positives due to the absence of a rider counting mechanism. In situations with relatively congested roads, Euclidean distance-based rider detection [47] using rider bounding boxes is ineffective and lacks qualitative analysis on a sizable dataset. YOLOv4 and Curriculum Learning (CL) based motorcycle riding violation detector [48] using a database generated from dynamic dash cameras was developed. CL approach is employed to tackle occlusions and trapezium rider boundary detector increased rider and motorcycle relevance in traffic scenes producing 84.44, 73.07, and 78.34% triple riding precision, recall, and F1-scores. Since the dataset generated contains traffic images captured at low heights, the system will not perform well when tested on images captured from real traffic junction poles. YOLO v8 with a Depth estimation algorithm [6] for identifying two-wheelers with triple riders was developed on a university campus dataset. However, the model demonstrated its effectiveness in low-traffic areas where people are not present near the vehicle's vicinity. The dataset contains images captured in direct line of sight of vehicles and systems performance heavily depends on height as well as capture angle. This doesn't represent real-world scenarios. On the COCO dataset, the YOLOv8-based Traffic Signal Violation Detection System [49] achieved 76% accuracy. Low illumination, distorted photos, and bad weather all affect the system's accuracy.

Sufficient work has been carried out in vehicle and helmet violation detection research. Triple-riding detection in constrained environments remains an open challenge. Sufficient datasets depicting real-world traffic scenes are needed to develop a better learning model and determine the violations more accurately, including occluded objects.

3 Proposed system

The proposed system utilizes the video input from the cameras mounted at various heights and detects triple rider violations using DetectNet_v2, and YOLOv8 models. The system implementation involves numerous essential steps which are illustrated in Fig. 5. The key steps are as follows: image acquisition and annotation, rider prediction after motorcycle and rider localization, rider recognition, triple-riding violation detection, and number plate information extraction. To begin with, the image acquisition and annotation process involves gathering diverse traffic data from traffic intersections and carefully annotating the images. The resulting ground truth data helps the model to learn accurately. This dataset is pre-processed to enhance the quality and serves as the foundation for training the detection model. Next, the motorcycle localization followed by accurate rider prediction is carried out by a trained DetectNet_v2 model. The trained YOLOv8 model is responsible for detecting the number of riders on a two-wheeler and recognizing the number plate. The final step involves triple-riding violation detection and violator identification through the number plate information extraction process. A detailed explanation of the above-mentioned steps is provided in the following sub-sections.

3.1 Dataset preparation

This process begins by extracting videos from 2 MP PTZ (Pan, Tilt, and Zoom) and 5 MP bullet cameras placed at heights between 10 and 14 feet. PTZ and bullet surveillance are popular because of wide-angle capture (180–360°) and night vision capabilities.

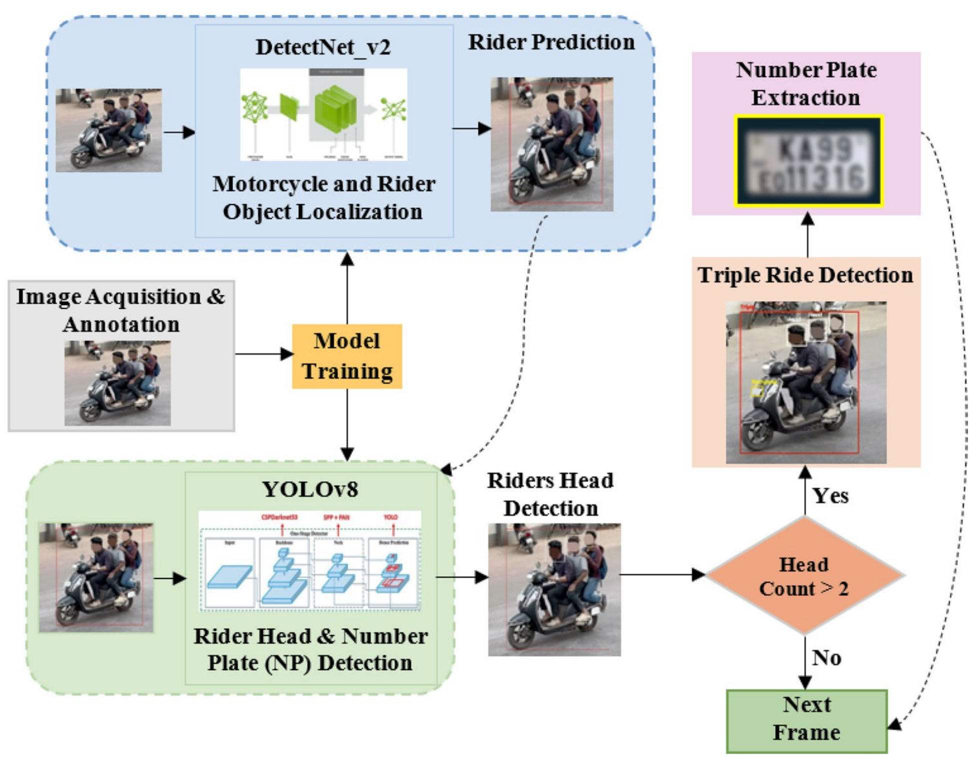


Fig. 5 Workflow of the proposed system

Hence, they are popularly deployed in traffic management systems to offer real-time information on traffic flow and incidents. Image extraction from live video feed comprises of the following key steps:

i. **Video Processing**: In this step, real-time video feed from the above-mentioned cameras is captured.
ii. **Frame Identification**: We extract, and preprocess the frames from the previous stage, and feed them into deep learning models. The frames are extracted at 10 frames per second (fps) rate.

In order to create an extensive data set for our study, we gathered 6564 images in total. These images were produced by extracting frames from videos and creating standalone images using advanced synthesis techniques. To ensure that these images closely reflect real-world traffic conditions, the videos were meticulously created to mimic real-world situations. The following factors were considered when creating the dataset:

Various Heights and Angles: To replicate the precise positions of real traffic cameras, cameras were positioned at different heights (10–14 feet) and angles (180–360°). This was done to make sure that various kinds of real-world camera positions and perspectives could be included in the dataset.

Day and Night Conditions: We made sure the dataset could accurately depict the variations in visibility and look of traffic scenarios during the day and at night by gathering data under various lighting conditions.

We carried out a thorough cleaning procedure to guarantee data quality and relevancy. As a result, 2551 good-quality images containing single, double, and triple rider images were added to the dataset. We further added 654 images of different numbers of riders from the online COCO dataset resulting in the final dataset number of 3205 image samples.

Table 1 highlights the breakdown of these images. Figure 6 displays the dataset that was utilized to train and test our model. Figure 6a illustrates sample images produced by placing cameras at multiple locations, whereas Fig. 6b presents sample images selected from internet sources like the COCO dataset. We preprocess these images to make them suitable for further processing in the next step.

3.1.1 Dataset preprocessing

A crucial stage in preparing datasets for machine learning applications like triple riding detection is data cleaning, also known as preprocessing. To make sure the dataset is appropriate, reliable, and useful for training and assessing models, this procedure consists of multiple steps.

These are the steps for data refining.

i. **Image Quality Assessment**: We identify frames having blurriness, poor lighting, occlusions, or irrelevant content. Images that do not meet quality or duplicate frames are discarded. Of 6564 raw frames, 2551 were selected for labelling and annotation jobs.
ii. **Image Annotation**: We set out to annotate our dataset using the Make Sense AI platform, which is well-known for its accuracy and efficiency in computer-vision-based image annotation tasks for background segmentation, object detection, and classification. From delineating an object's region-of-interest and outlining regions requiring segmentation, Make Sense AI provided us with the arsenal of tools necessary to achieve precise and accurate annotations. To handle cases where multiple instances of triple riding occur in the same image by accurately labelling each individual involved. Properly delineate overlapping instances to provide comprehensive annotations. Images' annotated metadata is first saved as.xml files (PASCAL VOC format), and then it is transferred to KITTI file format. This typically includes the location and characteristics of the motorcycle objects within the images such as their bounding boxes (xmin, ymin, xmax, ymax) labels and their attributes. Following successful training, DetectNet

Table 1 Custom dataset classification

Dataset source	Triple riding	Double riding	Single riding	Total images
Acquired images	2010	263	278	2551
Online dataset (COCO)	111	291	252	654
Class-wise total images	2121	554	530	3205

Fig. 6 Sample images used in generating the dataset **a** custom dataset, **b** online source

outputs the identified rider two-wheeler regions as bounding box coordinates. This information serves as the input for the subsequent stage.

3.2 Model selection

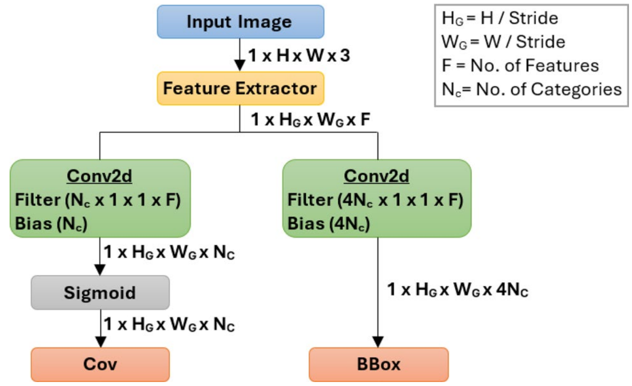
Following the preparation of the data set, the DetectNet_v2 is trained so that precise two-wheeler with rider information can be passed onto the next model. We train the YOLOv8 so that triple-riding violations can be predicted, and riders can be punished based on the processed number plate information.

3.2.1 DetectNet_v2 for two-wheeler and rider prediction

DetectNet_v2 is one of the model architectures that may be used to create Object Detection models using NVIDIA's Train Adapt Optimize (TAO) toolbox. TAO uses a transfer learning approach to build optimal Machine Learning (ML) models from smaller datasets. DetectNet_v2 supports Resnet10/18/34/50, VGG16/19, Googlenet, mobilenet_v1/_v2, and DarkNet19/53 feature extractor model types. Additionally, pre-trained models for each of these feature extractors are included. A portion of the Google OpenImages dataset was used to train these pre-trained models. As seen in Fig. 7, DetectNet_v2 employs a GridBox architecture to separate the input into a consistent grid and regression technique to predict a bounding box (bbox) and confidence value (cov) for each category. To predict four normalized bounding-box parameters i.e. "xc", "yc", "w", and "h" as well as the confidence value for each output class, the Gridbox system splits an input image into a grid. Once the model's output has been scaled and offset is defined, post-processing methods such as Non-Maximum Suppression (NMS) clustering are required to produce the final bounding boxes and class labels. Each bounding box in the image is assigned a confidence score by the DetectNet_v2, which also calculates the confidence that the bounding box encloses an object and forecasts the likelihood of belonging to its class. Equation 1's confidence score shows how closely the predicted bounding box resembles an actual object in the image.

$$\text{Confidence} = \Pr(\text{Object}) * \text{IoU} \tag{1}$$

Fig. 7 DetectNet_v2 high-level architecture



Intersection of Union (IoU) represents the intersection of 2 bounding-boxes over their union. It is obtained using,

$$\text{IOU} = \text{Areaof Intersection of boxes} / \text{Area of Union of two boxes} \tag{2}$$

The bounding box that has the highest level of confidence is kept, while the others are ignored.
The model is tested with the following parameters:

Input image (W × H × C) = 960 × 544 × 3.
H = 544 (Image Height), W = 960 (Image Width), and C = 3 (Channel).
Input Channel Ordering (N × C × H × W) = Where N = 8 (Batch Size).
Output: Confidence Scores (Floating point values), Bounding Box Coordinates (X, Y), Width (W), Height (H), and Labels (Text).
Occlusion: Objects with less than 60% readability are left unlabelled, while those with a bounding box surrounding the viewable area are marked as partially occluded.

The performance of the proposed rider two-wheeler detection is illustrated in Fig. 8. Despite multiple vehicles in the background, the blue bounding box shows accurate two-wheeler detection, while the white box shows the rider detection on a two-wheeler.

3.2.2 YOLOv8 triple-riding detection and rider identification

In 2023, Ultralytics[1] launched YOLOv8, the predecessor of the YOLO object detector family. It's the most advanced architectural design deployed for precise object detection, segmentation, and classification tasks. The two-wheeler rider image with bounding box values and labels produced by DetectNet_v2 are fed to this architecture to accurately detect and count rider heads and extract the owner information through the OCR number plate extraction method integrated into YOLOv8 architecture. The description of YOLOv8 architecture for triple-riding detection and rider identification is illustrated in Fig. 9.

Fig. 8 DetectNet_v2 rider two-wheeler detection output indicated by the blue bounding box

[1] Ultralytics YOLOv8, https://docs.ultralytics.com/models/yolov8/.

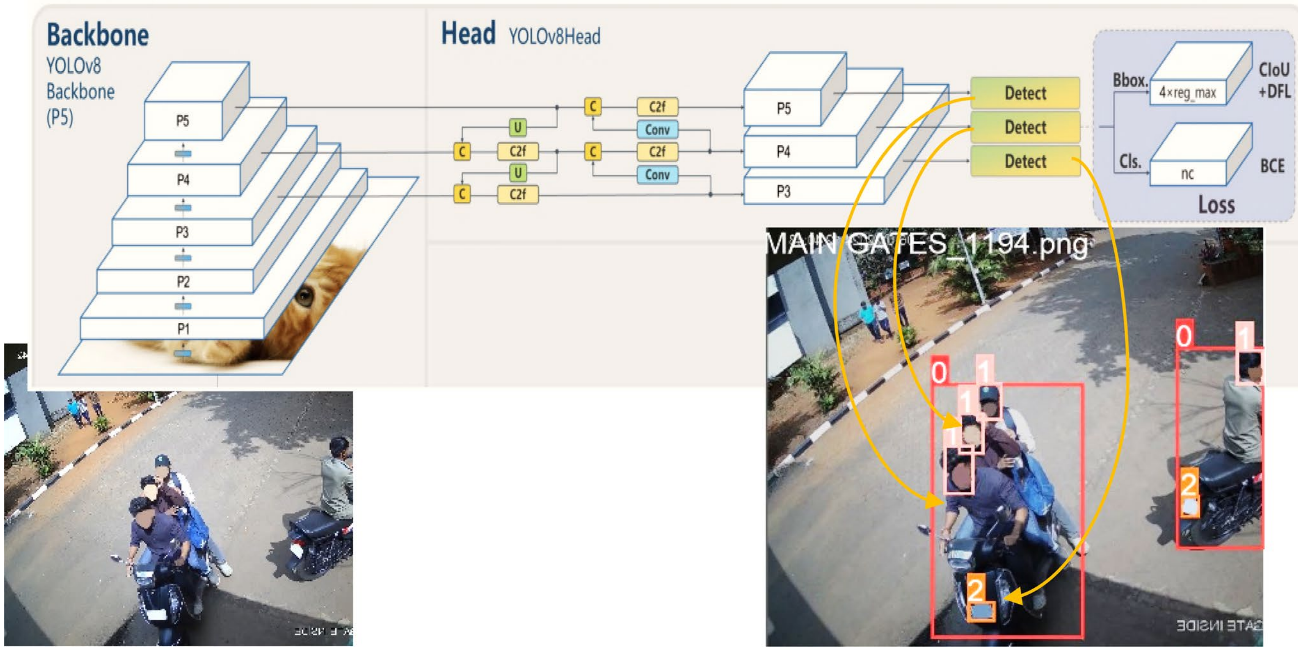


Fig. 9 YOLOv8 architecture for triple-riding detection and rider identification [9]

Input: The mosaic data augmentation can explore its crucial function in improving computer vision models' performance. Firstly, input images are refined using Mosaic augmentation, which transforms training by combining many images and increases the model's resilience and generalization abilities.

Backbone: The architecture uses a local feature analysis technique rather than reviewing the entire image; the goal of this strategy is primarily to reduce processing effort and enable real-time detection. The approach uses convolutions many times to extract feature maps.

Given probability distributions (p)X(x) and (p)Y(y), the convolution process can be thought of as a weighted average or sum of random occurrences. The convolution of X and Y is described by Eq. 3.

$$(X * Y)(z) = \int_{-\infty}^{\infty} pX(x) \cdot pY(z - x)dx \tag{3}$$

Darknet53, a novel CNN architecture with a focus on feature extraction and simplified filter windows and residual connections, forms the foundation of this framework. By using Cross Stage Partial (CSP) connections, the architecture can reduce computation and produce a richer gradient flow.

Neck: Its goal is to integrate the head and backbone network to enhance the features that the backbone retrieves, with a focus on enhancing spatial and semantic information at different scales. By removing the network's fixed-size constraint, a Spatial Pyramid Pooling (SPP) module eliminates the requirement for image warping, augmentation, or cropping. A CSP-Path Aggregation Network module comes next, which shortens the information path between lower and upper levels by incorporating the features that were learned in the backbone.

Head: It has three branches, each of which provides a distinct feature scale. Three grid cells (13×13, 26×26, 52×52) are utilized so that each of them predicts three bounding boxes to generate a bounding box, class probabilities, and confidence scores. Finally, the network uses NMS to filter out overlapping bounding boxes. Anchor boxes are fixed-sized sized bounding-boxes that are used to forecast the size and position of objects within an image. Instead of predicting random bounding boxes for each object instance, the model uses predefined aspect ratios and scales to forecast the anchor box coordinates, which are then modified to fit the object instance.

The neural network design makes use of both the Path Aggregation Network (PAN) and Feature Pyramid Network (FPN) where FPN gradually reduces the spatial resolution of the input image with the rise in the number of feature channels. As a result, feature maps that can recognize objects at different scales and resolutions are generated. The PAN architecture, on the other hand, integrates components from many network levels through skip connections. To accurately detect objects of different shapes and sizes, the network will be better equipped to record features at different scales and resolutions.

1. The YOLOv8 model achieves a high confidence score by using target regression on the spatially separated target bounding box. The detection process is summarized below:
2. The model divides the image into convolutions of size N x N (e.g., 13 × 13, 26 × 26, 52 × 52), with each cell's size determined by the input image.
3. Each of the N x N cells is in charge of determining the number of bounding boxes in the input image.
4. For each box in the input, the deconvolutional network predicts the bounding box's confidence in including the object, as well as the likelihood that the enclosed object belongs to one of the classes indicated in the configuration file.
5. Following that, it uses NMS to remove bounding boxes with a lower confidence value.
6. The processed images are subjected to the IOU function (Eq. 2), which was developed to determine the relationship between the riders observed in the frame and the two-wheeler.

As a result, the YOLOv8 model accurately recognizes rider on a two-wheeler (label 0—red bounding box), identifies if the number of riders on that two-wheeler is greater than two (label 1—pink bounding box), then extracts number plate information (label 2—orange bounding box) as shown Figs. 9 and 10.

3.2.3 Number plate recognition

Our final stage of the research is to integrate the trained Automatic Number Plate Recognition (ANPR) model into a triple-riding detection system and deploy it for real-time violation detection. We developed an algorithm that encompasses the

Fig. 10 Triple-riding detection and rider identification instances with the following information: label 0 (red box)—two-wheeler, label 1 (pink box)—triple rider heads, label 2 (orange box)—number plate

full ANPR process, for recognizing and interpreting the individual numbers on the retrieved number plate and display it on the interface as seen in Fig. 11.

4 Performance evaluation

As discussed in Sect. 3.1, we use our custom dataset of 3205 images containing 2121 triple riding, 554 double riding, and 530 single riding image instances. We use the 80:20 split for training and testing our data. The object detection model's effectiveness is determined by the following parameters:

i. **Intersection over Union (IOU)**—First, the intersection of two boxes for the same object is calculated to find the IOU between the ground truth and predicted boundary boxes. The Union determines the overall area covered by the boundary boxes, while the intersection defines the overlap. The predicted boundary box's accuracy in detecting the item is measured using Eq. 2.

ii. **Mean Average-Precision (mAP)**: The weighted average precision at each threshold level is represented as average precision, whereas the average precision for each class is averaged by mAP, which is calculated using the following formula.

$$\mathrm{mAp} = 1/n \sum_{i=1}^{n} AP_i \tag{4}$$

Box-loss (train/val box_loss): A rider bounding-box mapping between the ground truth and predicted objects is measured here. Lower values indicate better performance.

iv. **Training class loss (train/val cls_loss**: These metrics assess how well the algorithm can identify riders in particular regions of an image. Lower values indicate better performance.

v. **Training distortion loss (train/val dfl_loss)**: It refers to the loss caused by the model distortion. It assesses how accurately the model forecasts the rider's position. Lower values indicate better performance.

vi. **Metrics/precision (B)**: This indicates the rider class's precision that counts the actual riders among the riders detected by the model. Higher values indicate better performance.

vii. **Metrics/recall (B)**: Indicates the rider class's recall. The number of real two-wheelers in the frame that the model identified correctly is referred to as recall. Higher values indicate better performance.

viii. **Metrics/mAP50 (B)**: It is the mAP at 50% IOU and is frequently used to assess the performance of object-detecting systems. A single score representing overall object detection ability is produced by precision and recall values. A higher value suggests improved performance.

ix. **Metrics/mAP50-95 (B)**: This metric is mAP at 95 and 50% IOU levels meaning that it evaluates the model's performance across a wider range of IOU thresholds, making it similar to the mAP50 (B). A higher value suggests improved performance.

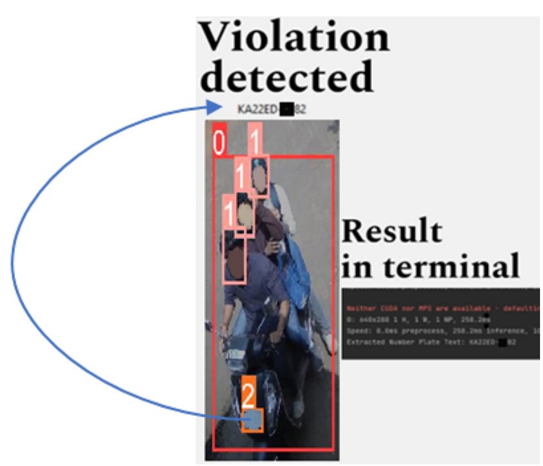


Fig. 11 ANPR triple-riding violators number plate recognition on GUI

4.1 YOLOv8 triple-riding detection training results

YOLOv8 is responsible for identifying and distinguishing two-wheeler riders from other vehicles in the traffic scene. This step takes the use of a carefully selected dataset of two-wheeler riders in a variety of traffic circumstances. The model is being trained to identify riders and create bounding boxes around them. Following successful training, the model outputs the detected two-wheelers along with riders as bounding box coordinates or cropped pictures. This data is used for triple-riding detection and number plate extraction to identify triple-riding violating riders. Loss curves often decline as the model is trained. This indicates that the model is acquiring information. This is a common behaviour in machine learning that indicates the likelihood of the model overfitting the training data. In both the training and validation sets, the precision and recall scores appear to be greater than 0.5. This demonstrates how well the model recognizes motorcycle riders in images. Additionally, in both the training and validation sets, the mAP metrics are higher than 0.5. This offers more proof of the model's excellent performance. The curves in Fig. 12 represent how well our YOLOv8 network is learning to recognize riders from videos.

4.1.1 Recall-confidence plot

For riders on two-wheelers (riders), triple-riders (heads), and number plates (number plates) classes, this plot shows how the recall metric varies as the confidence threshold changes (Fig. 13).

It depicts recall on the y-axis, which represents the model's accuracy in recognizing objects in images (such as motorcycle riders, triple-riders, and license plates). The model's confidence in its predictions is shown by the x-axis confidence-score. Despite having a low confidence score, the model has a high recall for two-wheeler riders, as seen by the "rider" curve. This implies that the model can recognize many riders on two-wheelers in traffic images, regardless of the prediction's accuracy. The recall remains strong as the confidence level climbs, plateauing and then slightly dropping. This means that even when the model is unsure about its prediction, it is still quite good at detecting "triple-riders" in images.

The "numberplate" curve is similar to the "triple-riders". Despite poor confidence scores, the model has a good recall rate for license plates. As the confidence score improves, so does the recall value, but it finally plateaus and drops. This means that the model can also identify license plates in photos. The "triple-rider" curve is unique from the other two curves. At lower confidence levels, it starts much lower and progressively grows as confidence increases.

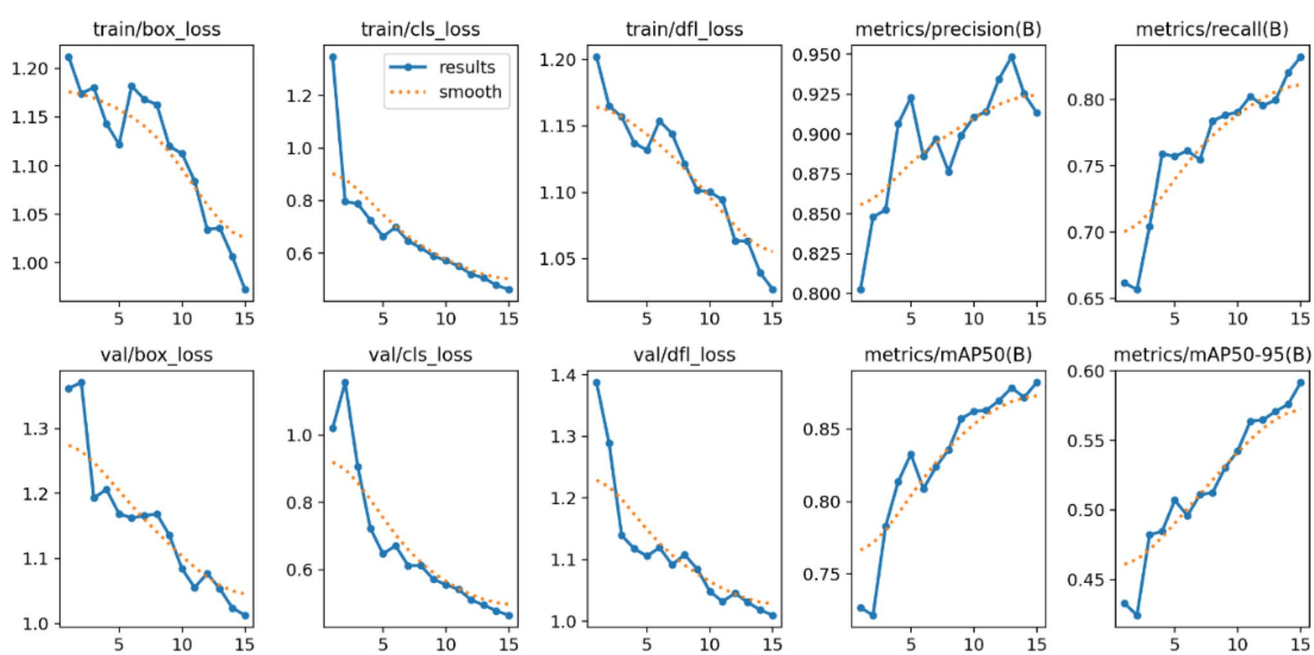


Fig. 12 YOLOv8 triple riding detection models training/validation curves and performance parameters

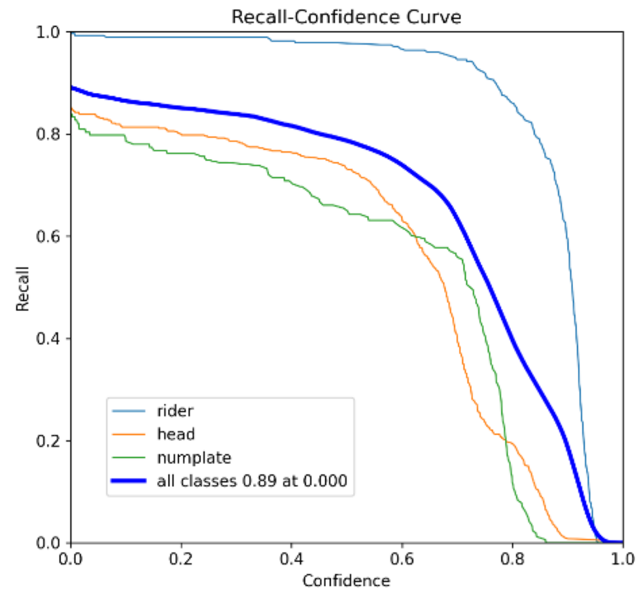


Fig. 13 YOLOv8 triple riding and number plate detection recall-confidence curve

4.1.2 Precision-recall plot

The detection accuracy of riders on two-wheelers (riders), triple-riders (heads), and number plates (number plates) is tested using a precision-recall curve, as shown in Fig. 14. Precision (y-axis) depicts the percentage of correctly detected positives. It refers to the percentage of times the model correctly identified an object (a rider on a two-wheeler, a triple-rider, or a number plate) in the inputted image. On the other hand, the recall (x-axis) shows the proportion of real positive samples that the model discovered. The percentage of actual two-wheelers with riders, triple riders, and license plates that the model accurately identified in a particular image is referred to by this precision-recall plot. The curves for each class, namely two-wheelers (riders), triple-riders (heads), and numberplates, provide the following information: The model can accurately recognize riders on two-wheelers even at lower thresholds, as indicated by the "rider" curve, which starts high on the precision and recall axes. The precision remains high, but recall begins to drop as the precision threshold rises (the model becomes more severe in recognizing something as a two-wheeler rider). This means that, while the model may be effective at recognizing most two-wheeler riders, it may miss others since it prioritizes constructing more precise classifications.

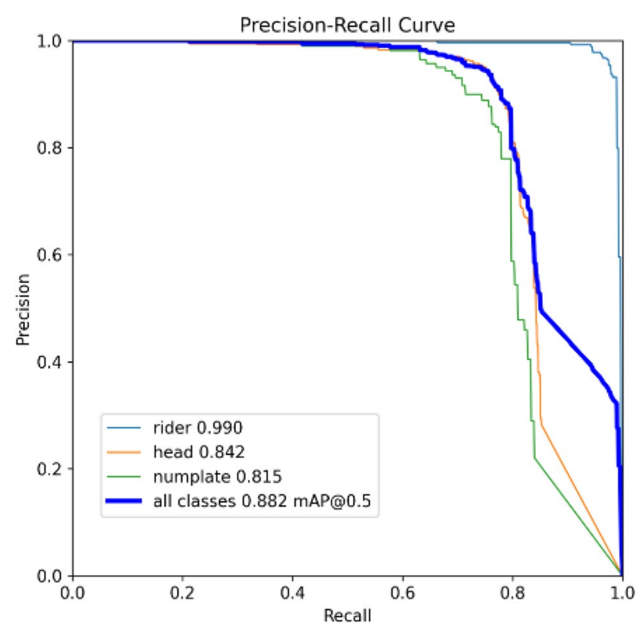


Fig. 14 YOLOv8 triple riding and number plate detection precision-recall curve

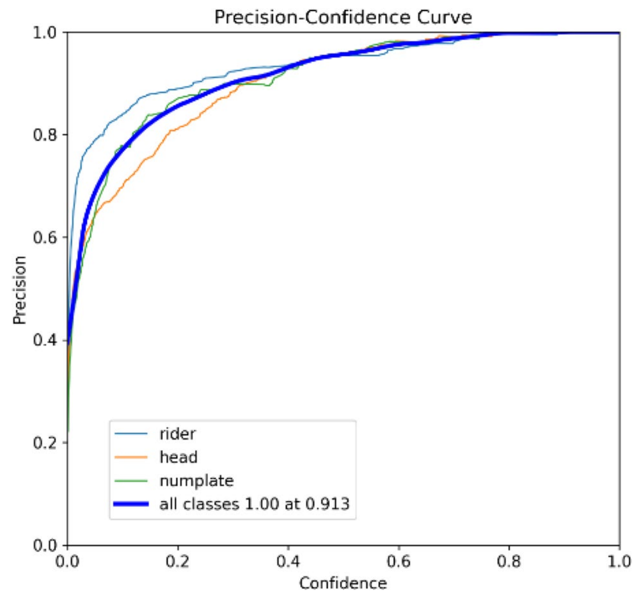


Fig. 15 YOLOv8 triple riding and number plate detection precision—confidence curve

Table 2 Confusion matrix indicating triple riding detection performance

N = 641		Actual	
		Triple riding violation	No violation
Predicted	Triple riding violation	239 (TP)	17 (FP)
	No violation	38 (FN)	347 (TN)

The "numberplate" curve looks similar to the rider curve. At lower thresholds, the model's precision and recall are good; at higher thresholds, the precision is great, but the recall decreases slightly. According to this, the model does an adequate job of identifying license plates, with the same trade-off between recall and precision as two-wheeler riders. The curve for "triple-riders (heads)" deviates significantly from the other two. As recall increases, precision gradually improves from its initial low level. This suggests that the model performs poorly at lower thresholds (more likely to produce False Positives) and improves as the threshold is raised (more accurate but may miss some true cases).

4.1.3 Precision-confidence plot

The accuracy of the model for various object classes at varying confidence levels is shown in the precision-confidence curve of Fig. 15. For a given prediction's accuracy, the model's confidence level is shown on the graph's x-axis. The precision is shown on the Y axis of the graph since it is computed by splitting the percentage of correct positive classifications (True Positives) by the total number of predicted positive classifications (True Positives as well as False Positives). Figure 15 shows three curves representing the accuracy of three different object classes: numberplate, rider on a two-wheeler (rider), and triple-riders (head). The model has an accuracy of 1.0 (or 100%) for all classes at a confidence level of 0.913. When the confidence level is 0.913 or higher, it means that the model is quite confident in its predictions for all object classes.

4.1.4 Confusion matrix

Table 2 describes the confusion matrix for a YOLOv8 model's performance on a triple-riding detection task. The triple-riding experiment's performance is evaluated using several performance metrics. Table 3 lists the performance metrics and their values for our proposed (DetectNet_v2 + YOLOv8) vs. the YOLOv8 [6], and YOLOv3 [7] models.

Table 3 compares the performance of the proposed model with previous state-of-arts. Our proposed (DetectNet_v2 + YOLOv8) system produced accuracy, precision, recall, and F-score of 91.42%, 93.4%, 86.3%, and 90% respectively. The model's efficiency is evaluated using the mAP50 (IOU at 0.50 threshold) and mAP50-90 (average precision for

Table 3 Triple-riding experimental results and performance comparison with previous stat-of-art models

Performance metrics	Derivations	Scores		
		Proposed (DetectNet_v2 + YOLOv8)	YOLOv8 [6]	YOLOv3 [7]
Accuracy	ACC = (TP + TN)/(P + N)	0.914	0.91	0.917
Precision	PREC = TP/(TP + FP)	0.934	0.889	0.9
Sensitivity (recall)	TPR = TP/(TP + FN)	0.863	0.914	1
F-score	F1 = 2*TP/2*TP + FP + FN	0.9	0.901	0.947
False-positive rate	FPR = FP/(FP + TN)	0.047	–	–
Specificity	SPC = TN/(FP + TN)	0.953	–	–
mAP50	–	0.842	0.938	0.90
mAP50-90	–	0.586	0.578	–

IOU threshold ranges from 50 to 90, with a 5 increment). The proposed system generated mAP50 and mAP50-90 of 84.2%, and 58.6%, respectively. The results reveal that the proposed (DetectNet_v2 + YOLOv8) framework leads to higher mean average precision for triple-riding violation detection highlighting its advantages over the YOLOv8 [6] and YOLOv3 [7] models. The model's improved two-wheeler detection and rider identification by DetectNetv2 helped to lower the False rates.

As discussed in Sect. 1, the dataset used to train and validate YOLOv8 [6] and YOLOv3 [7] models was captured from near view and lower camera angles. These datasets do not represent the actual traffic scenarios and the model's performance heavily depends on the camera position, angles, and absence of pedestrians around the vehicle. Our proposed model ignores the pedestrians on the road when detecting riders on a two-wheeler and it demonstrated the ability to perform well even when the camera is placed at different heights and angles. In complex traffic scenarios, handling motorcycle rider occlusions, and false rider predictions by our model is still challenging. The model's performance is affected due to poor lighting conditions due to bad weather or night conditions. Additionally, our model generated 98% and 81% accuracy rates for two-wheeler rider detection and numberplates respectively.

5 Conclusion and future scope

One of the most difficult challenges faced by real-time traffic violation detection systems is correctly detecting two-wheeler riders in complex traffic settings. We have proposed a deep learning-based automatic object detection system to correctly identify and penalize triple-riding riders from cameras placed at various heights and distances. The proposed DetectNet_v2 and YOLOv8 models perform automatic real-time two-wheeler rider localization, triple-riding violation detection, and number plate extraction. The model is tested on 641 images and the accuracy level of 91.42, 98, and 81% for a triple rider, numberplate, and rider with two-wheeler detection is achieved. Our solution works well without any manual intervention in the triple-riding detection process and is robust against the placement of a camera position or pedestrians on the road. This system can be integrated with the existing traffic management and surveillance cameras to monitor triple-riding traffic offenses in real-time.

However, there are major scaling issues with implementing automatic traffic violation detection especially in managing massive amounts of video data from several camera feeds and satisfying the precise violation identification and detections through real-time image processing. The edge AI solutions provide powerful AI processing on the edge, and they don't require a continuous cloud connection. The integration of multiple traffic cameras with edge AI devices can be explored to overcome the above problems. The existing traffic violation systems suffer while handling footage collected from a variety of weather conditions (bad weather, poor illumination, and occlusions). To overcome these challenges, an ensemble-based deep learning system with variable hyperparameters can be developed. By creating a synthetic dataset for various traffic situations, our suggested approach can be extended to identify more traffic infractions, such as helmet violations, wrong-side driving, and Overspeed identification, thereby strengthening the traffic department.

Acknowledgements We acknowledge the Founder and CEO of DocketRun Technologies Pvt. Ltd., Hubbali, Karnataka, India, for identifying the research problem and guiding the development of research solutions.

Author contributions Authors Uttam U. Deshpande, Kevin Pinto, Arjun J, Mayankraj Patil, and Balagouda Nirwani were involved in data collection, model design, experimentation, interpretation of results, and research project supervision. Authors Uttam U. Deshpande, Supriya Shanbhag, Rudragoud Patil, Ramchandra Alias Ameet Chate, Sufola Das Chagas Silva Araujo were involved in study conception, survey study and draft manuscript preparation.

Funding Funding Information is not applicable.

Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participation Not Applicable.

Consent to Publication Not Applicable.

Competing interests The authors declare that they have no conflicts of interest.

References

1. Road Accidents in India 2022 published by Ministry of Road Transport and Highways, Transport research wing, New Delhi. Government of India. [Last accessed on 2025 Feb 19]. Available from https://morth.nic.in/sites/default/files/RA_2022_30_Oct.pdf.
2. Shangliao Sun, On-the-spot prosecutions made by Delhi traffic police for triple riding on two-wheelers from 2014 to 2022 [Internet]. [Last accessed on 2025 Feb 19]. Available from https://www.statista.com/statistics/1082087/india-triple-riding-prosecutions-delhi-traffic-police/.
3. Kaffash S, Nguyen AT, Zhu J. Big data algorithms and applications in intelligent transportation system: a review and bibliometric analysis. Int J Prod Econ. 2021;231:107868. ISSN 0925-5273. https://doi.org/10.1016/j.ijpe.2020.107868.
4. Chen C, Liu B, Wan S, Qiao P, Pei Q. An edge traffic flow detection scheme based on deep learning in an intelligent transportation system. IEEE Trans Intell Transp Syst. 2021;22(3):1840–52. https://doi.org/10.1109/TITS.2020.3025687.
5. Sun P, Boukerche A. AI-assisted data dissemination methods for supporting intelligent transportation systems†. Internet Technol Lett. 2021;4. https://doi.org/10.1002/itl2.169.
6. Chaturvedi P, Lavingia K, Raval G. Detection of traffic rule violation in University campus using deep learning model. Int J Syst Assur Eng Manag. 2023;14:2527–45. https://doi.org/10.1007/s13198-023-02107-8.
7. Mallela NC, Volety R, Ramalingam SP, Rk N. Detection of the triple riding and speed violation on two-wheelers using deep learning algorithms. Multimedia Tools Appl. 2021;80:1–13. https://doi.org/10.1007/s11042-020-10126-x.
8. Saumya A, Gayathri V, Venkateswaran K, Kale S, Sridhar N. Machine learning based surveillance system for detection of bike riders without helmet and triple rides. In: 2020 International Conference on Smart Electronics and Communication (ICOSEC), Trichy, India; 2020. p. 347–52. https://doi.org/10.1109/ICOSEC49089.2020.9215266.
9. Object Detection using TAO DetectNet_v2, [Internet]. [Last accessed on 2025 Feb 19]. Available from https://docs.nvidia.com/tao/tao-toolkit/text/cv_finetuning/tensorflow_1/object_detection/detectnet_v2.html.
10. Chairat A, Dailey MN, Limsoonthrakul S, Ekpanyapong M, Raj KCD. Low cost, high performance automatic motorcycle helmet violation detection. In: 2020 IEEE Winter Conference on Applications of Computer Vision (WACV), Snowmass, CO, USA; 2020. p. 3549–3557. https://doi.org/10.1109/WACV45572.2020.9093538.
11. Khan FA, Nagori N, Naik A. Helmet and number plate detection of motorcyclists using deep learning and advanced machine vision techniques. In: 2020 Second International Conference on Inventive Research in Computing Applications (ICIRCA), Coimbatore, India; 2020. p. 714–7. https://doi.org/10.1109/ICIRCA48905.2020.9183287.
12. Shine L, Jiji CV. Automated detection of helmet on motorcyclists from traffic surveillance videos: a comparative analysis using hand-crafted features and CNN. Multimedia Tools Appl. 2020;79(19–20):14179–99. https://doi.org/10.1007/s11042-020-08627-w.
13. Mehmood M, Shahzad A, Zafar B, Shabbir A, Ali N. Remote sensing image classification: a comprehensive review and applications. Math Probl Eng. 2022;2022:5880959. https://www.hindawi.com/journals/mpe/2022/5880959/
14. Yang M. Research on vehicle automatic driving target perception technology based on improved MSRPN algorithm. J Comput Cogn Eng. 2022;1(3):147–51. https://doi.org/10.47852/bonviewJCCE20514.

15. Ali M, Asghar MZ, Shah M, et al. A simple and effective sub-image separation method. Multimed Tools Appl. 2022;81:14893–910. https://doi.org/10.1007/s11042-021-11680-8.
16. Ashiq F, Asif M, Ahmad M, Zafar S, Massod K, Mahmood T, Mahmood M, Lee Ik. CNN-based object recognition and tracking system to assist visually impaired people. IEEE Access. 2022;10:1–1. https://doi.org/10.1109/ACCESS.2022.3148036.
17. Yang Y, Song X. Research on face intelligent perception technology integrating deep learning under different illumination intensities. J Comput Cogn Eng. 2022;1(1):32–6. https://doi.org/10.47852/bonviewJCCE19919.
18. Uzma, Al-Obeidat F, Tubaishat A et al. Gene encoder: a feature selection technique through unsupervised deep learning-based clustering for large gene expression data. Neural Comput Applic. 2022;34:8309–31. https://doi.org/10.1007/s00521-020-05101-4.
19. Haq HB, Asif M, Ahmad M, Ashraf R, Mahmood T. An effective video summarization framework based on the object of interest using deep learning. Math Probl Eng. 2022;2022:1–25. https://doi.org/10.1155/2022/7453744.
20. Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition; 2014. arXiv:1409.1556.
21. Girshick R, Donahue J, Darrell T, Malik J. Region-based convolutional networks for accurate object detection and segmentation. IEEE Trans Pattern Analy Mach Intell. 2016;38(1):142–58. https://doi.org/10.1109/TPAMI.2015.2437384.
22. Szegedy C et al. Going deeper with convolutions. In: 2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Boston, MA, USA; 2015. p. 1–9. https://doi.org/10.1109/CVPR.2015.7298594.
23. Krizhevsky A, Sutskever I, Hinton GE. ImageNet classification with deep convolutional neural networks. Commun ACM, 2017;60(6):84–90. https://doi.org/10.1145/3065386.
24. Lohia A, Kadam KD, Joshi RR, Bongale AM. Bibliometric analysis of one-stage & two-stage object detection. Libr Philos Pract. 2021;4910:34.
25. Girshick R, Donahue J, Darrell T, Malik J. Rich feature hierarchies for accurate object detection and semantic segmentation. In: 2014 IEEE Conference on Computer Vision and Pattern Recognition, Columbus, OH, USA; 2014. p. 580–7. https://doi.org/10.1109/CVPR.2014.81.
26. Dai J, Li Y, He K, Sun J. R-FCN: object detection via region-based fully convolutional networks. In: Proceedings of the 30th International Conference on Neural Information Processing Systems (NIPS'16). Curran Associates Inc., Red Hook, NY, USA; 2016. p. 379–87.
27. Ren S, He K, Girshick R, Sun J. Faster R-CNN: towards real-time object detection with region proposal networks. IEEE Trans Pattern Anal Mach Intell. 2017;39(6):1137–49. https://doi.org/10.1109/TPAMI.2016.2577031.
28. Girshick R. Fast R-CNN. In: 2015 IEEE International Conference on Computer Vision (ICCV), Santiago, Chile; 2015. p. 1440–48. https://doi.org/10.1109/ICCV.2015.169.
29. Liu W, Anguelov D, Erhan D, Szegedy C, Reed S, Fu C-Y, Berg A. SSD: single shot multibox detector. 2016;9905:21–37. https://doi.org/10.1007/978-3-319-46448-0_2.
30. Redmon J, Divvala S, Girshick R, Farhadi A. You only look once: unified, real-time object detection. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA; 2016. p. 779–88. https://doi.org/10.1109/CVPR.2016.91.
31. He K, Gkioxari G, Dollar P, Girshick R. Mask R-CNN. IEEE Trans Pattern Anal Mach Intell. 2020;42(2):386–97. https://doi.org/10.1109/TPAMI.2018.2844175.
32. Xing H, Xiao Z, Qu R, Zhu Z, Zhao B. An efficient federated distillation learning system for multitask time series classification. IEEE Trans Instrum Measur. 2022;71:1–12. https://doi.org/10.1109/TIM.2022.3201203.
33. Bochkovskiy A, Wang C-Y, Liao H. YOLOv4: optimal speed and accuracy of object detection; 2020. https://doi.org/10.48550/arXiv.2004.10934.
34. Redmon J, Farhadi A. YOLOv3: an incremental improvement ; 2018. https://doi.org/10.48550/arXiv.1804.02767.
35. Wang H, Yu Y, Cai Y, Chen L, Chen X. A vehicle recognition algorithm based on deep transfer learning with a multiple feature subspace distribution. Sensors. 2018;18:4109. https://doi.org/10.3390/s18124109.
36. Desai GG, Bartakke PP. Real-time implementation of Indian license plate recognition system. In: 2018 IEEE Punecon, Pune, India; 2018. p. 1–5.https://doi.org/10.1109/PUNECON.2018.8745419.
37. Afzal A, Draz HU, Khan MZ, Khan MUG. Automatic helmet violation detection of motorcyclists from surveillance videos using deep learning approaches of computer vision. In: 2021 International Conference on Artificial Intelligence (ICAI), Islamabad, Pakistan; 2021. p. 252–7. https://doi.org/10.1109/ICAI52203.2021.9445206.
38. Kathane M, Abhang S, Jadhavar A, Joshi AD, Sawant ST. Traffic rule violation detection system: deep learning approach. In: Gupta D, Sambyo K, Prasad M, Agarwal S, editors. Advanced Machine Intelligence and Signal Processing. Lecture Notes in Electrical Engineering, vol. 858. Springer, Singapore; 202. https://doi.org/10.1007/978-981-19-0840-8_14.
39. Kumar S, Neware N, Jain A, Swain D, Singh P. Automatic helmet detection in real-time and surveillance video. In: Proceedings of ICMLIP; 2020. p. 51–60.
40. Yogameena B, Menaka K, Saravana Perumaal S. Deep learning-based helmet wear analysis of a motorcycle rider for intelligent surveillance system. IET Intelligent Transp Syst. 2019;13(7):1190–8. https://doi.org/10.1049/iet-its.2018.5241.
41. Jia W et al. Real-time automatic helmet detection of motorcyclists in urban traffic using improved YOLOv5 detector. IET Image Process. 2021;15:3623–37. https://doi.org/10.1049/ipr2.12295.
42. Lin H, Chen G, Siebert F. Positional encoding: improving class-imbalanced motorcycle helmet use classification. 2021;1194–8. https://doi.org/10.1109/ICIP42928.2021.9506178.
43. Rithish H, Modhugu R, Reddy R, Saluja R, Jawahar CV. Evaluating computer vision techniques for urban mobility on largescale, unconstrained roads. IEEE; 2021. https://doi.org/10.48550/arXiv.2109.05226.
44. Rajalakshmi NR, Saravanan K. Traffic violation invigilation using transfer learning. In: 2022 International Conference on Electronic Systems and Intelligent Computing (ICESIC), Chennai, India; 2022. p. 286–92. https://doi.org/10.1109/ICESIC53714.2022.9783519.
45. Agorku G, Agbobli D, Chowdhury V, Amankwah-Nkyi K, Ogungbire A, Lartey P, Aboah A. Real-time helmet violation detection using YOLOv5 and ensemble learning; 2023. https://doi.org/10.48550/arXiv.2304.09246.
46. Siebert FW, Lin H. Detecting motorcycle helmet use with deep learning. Accid Anal Prev. 2020;134: 105319. https://doi.org/10.1016/j.aap.2019.105319.

47. Goyal A, Agarwal D, Subramanian A, Jawahar CV, Sarvadevabhatla RK, Saluja R. Detecting, tracking and counting motorcycle rider traffic violations on unconstrained roads. In: 2022 IEEE/CVF conference on computer vision and pattern recognition workshops (CVPRW), New Orleans, LA, USA; 2022. p. 4302–11. https://doi.org/10.1109/CVPRW56347.2022.00476.
48. Gehani H. Traffic signal violation detection system using computer vision. J Electr Syst. 2024;20:2661–70. https://doi.org/10.52783/jes.2037.
49. Reis D, Kupec J, Hong J, Daoudi A. Real-time flying object detection with YOLOv8; 2023. https://arxiv.org/abs/2305.09972.